of $\mathcal{E}_X$. The constraints of the problem (5.14) are that $a_i \in \mathcal{E}_X$. Thus the problem (5.14) is to determine the minimum volume ellipsoid, centered at the origin, that includes the points $a_1, \ldots, a_m$.

The inequality constraints in problem (5.14) are affine; they can be expressed as

$$\mathbf{tr}\left((a_i a_i^T)X\right) \leq 1.$$

In example 3.23 (page 92) we found that the conjugate of f_0 is

$$f_0^*(Y) = \log\det(-Y)^{-1} - n,$$

with $\mathbf{dom}\, f_0^* = -\mathbf{S}_{++}^n$. Applying the result (5.11) above, the dual function for the problem (5.14) is given by

$$g(\lambda) = \begin{cases} \log\det\left(\sum_{i=1}^m \lambda_i a_i a_i^T\right) - \mathbf{1}^T\lambda + n & \sum_{i=1}^m \lambda_i a_i a_i^T \succ 0 \\ -\infty & \text{otherwise.} \end{cases} \tag{5.15}$$

Thus, for any $\lambda \succeq 0$ with $\sum_{i=1}^m \lambda_i a_i a_i^T \succ 0$, the number

$$\log\det\left(\sum_{i=1}^m \lambda_i a_i a_i^T\right) - \mathbf{1}^T\lambda + n$$

is a lower bound on the optimal value of the problem (5.14).

5.2 The Lagrange dual problem

For each pair (λ, ν) with $\lambda \succeq 0$, the Lagrange dual function gives us a lower bound on the optimal value $p^\star$ of the optimization problem (5.1). Thus we have a lower bound that depends on some parameters λ, ν. A natural question is: What is the *best* lower bound that can be obtained from the Lagrange dual function?

This leads to the optimization problem

$$\begin{array}{ll} \text{maximize} & g(\lambda, \nu) \\ \text{subject to} & \lambda \succeq 0. \end{array} \tag{5.16}$$

This problem is called the *Lagrange dual problem* associated with the problem (5.1). In this context the original problem (5.1) is sometimes called the *primal problem*. The term *dual feasible*, to describe a pair (λ, ν) with $\lambda \succeq 0$ and $g(\lambda, \nu) > -\infty$, now makes sense. It means, as the name implies, that (λ, ν) is feasible for the dual problem (5.16). We refer to $(\lambda^\star, \nu^\star)$ as *dual optimal* or *optimal Lagrange multipliers* if they are optimal for the problem (5.16).

The Lagrange dual problem (5.16) is a convex optimization problem, since the objective to be maximized is concave and the constraint is convex. This is the case whether or not the primal problem (5.1) is convex.

5.2.1 Making dual constraints explicit

The examples above show that it is not uncommon for the domain of the dual function,

$$\mathbf{dom}\, g = \{(\lambda, \nu) \mid g(\lambda, \nu) > -\infty\},$$

to have dimension smaller than $m + p$. In many cases we can identify the affine hull of $\mathbf{dom}\, g$, and describe it as a set of linear equality constraints. Roughly speaking, this means we can identify the equality constraints that are 'hidden' or 'implicit' in the objective g of the dual problem (5.16). In this case we can form an equivalent problem, in which these equality constraints are given explicitly as constraints. The following examples demonstrate this idea.

Lagrange dual of standard form LP

On page 219 we found that the Lagrange dual function for the standard form LP

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax = b \\ & x \succeq 0 \end{array} \tag{5.17}$$

is given by

$$g(\lambda, \nu) = \begin{cases} -b^T \nu & A^T \nu - \lambda + c = 0 \\ -\infty & \text{otherwise.} \end{cases}$$

Strictly speaking, the Lagrange dual problem of the standard form LP is to maximize this dual function g subject to $\lambda \succeq 0$, *i.e.*,

$$\begin{array}{ll} \text{maximize} & g(\lambda, \nu) = \begin{cases} -b^T \nu & A^T \nu - \lambda + c = 0 \\ -\infty & \text{otherwise} \end{cases} \\ \text{subject to} & \lambda \succeq 0. \end{array} \tag{5.18}$$

Here g is finite only when $A^T \nu - \lambda + c = 0$. We can form an equivalent problem by making these equality constraints explicit:

$$\begin{array}{ll} \text{maximize} & -b^T \nu \\ \text{subject to} & A^T \nu - \lambda + c = 0 \\ & \lambda \succeq 0. \end{array} \tag{5.19}$$

This problem, in turn, can be expressed as

$$\begin{array}{ll} \text{maximize} & -b^T \nu \\ \text{subject to} & A^T \nu + c \succeq 0, \end{array} \tag{5.20}$$

which is an LP in inequality form.

Note the subtle distinctions between these three problems. The Lagrange dual of the standard form LP (5.17) is the problem (5.18), which is equivalent to (but not the same as) the problems (5.19) and (5.20). With some abuse of terminology, we refer to the problem (5.19) or the problem (5.20) as the Lagrange dual of the standard form LP (5.17).

Lagrange dual of inequality form LP

In a similar way we can find the Lagrange dual problem of a linear program in inequality form

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax \preceq b. \end{array} \tag{5.21}$$

The Lagrangian is

$$L(x, \lambda) = c^T x + \lambda^T (Ax - b) = -b^T \lambda + (A^T \lambda + c)^T x,$$

so the dual function is

$$g(\lambda) = \inf_x L(x, \lambda) = -b^T \lambda + \inf_x (A^T \lambda + c)^T x.$$

The infimum of a linear function is $-\infty$, except in the special case when it is identically zero, so the dual function is

$$g(\lambda) = \begin{cases} -b^T \lambda & A^T \lambda + c = 0 \\ -\infty & \text{otherwise.} \end{cases}$$

The dual variable λ is dual feasible if $\lambda \succeq 0$ and $A^T \lambda + c = 0$.

The Lagrange dual of the LP (5.21) is to maximize g over all $\lambda \succeq 0$. Again we can reformulate this by explicitly including the dual feasibility conditions as constraints, as in

$$\begin{array}{ll} \text{maximize} & -b^T \lambda \\ \text{subject to} & A^T \lambda + c = 0 \\ & \lambda \succeq 0, \end{array} \tag{5.22}$$

which is an LP in standard form.

Note the interesting symmetry between the standard and inequality form LPs and their duals: The dual of a standard form LP is an LP with only inequality constraints, and vice versa. One can also verify that the Lagrange dual of (5.22) is (equivalent to) the primal problem (5.21).

5.2.2 Weak duality

The optimal value of the Lagrange dual problem, which we denote $d^\star$, is, by definition, the best lower bound on $p^\star$ that can be obtained from the Lagrange dual function. In particular, we have the simple but important inequality

$$d^\star \leq p^\star, \tag{5.23}$$

which holds even if the original problem is not convex. This property is called *weak duality*.

The weak duality inequality (5.23) holds when $d^\star$ and $p^\star$ are infinite. For example, if the primal problem is unbounded below, so that $p^\star = -\infty$, we must have $d^\star = -\infty$, *i.e.*, the Lagrange dual problem is infeasible. Conversely, if the dual problem is unbounded above, so that $d^\star = \infty$, we must have $p^\star = \infty$, *i.e.*, the primal problem is infeasible.

We refer to the difference $p^\star - d^\star$ as the *optimal duality gap* of the original problem, since it gives the gap between the optimal value of the primal problem and the best (*i.e.*, greatest) lower bound on it that can be obtained from the Lagrange dual function. The optimal duality gap is always nonnegative.

The bound (5.23) can sometimes be used to find a lower bound on the optimal value of a problem that is difficult to solve, since the dual problem is always convex, and in many cases can be solved efficiently, to find $d^\star$. As an example, consider the two-way partitioning problem (5.7) described on page 219. The dual problem is an SDP,

$$\begin{array}{ll} \text{maximize} & -\mathbf{1}^T\nu \\ \text{subject to} & W + \mathbf{diag}(\nu) \succeq 0, \end{array}$$

with variable $\nu \in \mathbf{R}^n$. This problem can be solved efficiently, even for relatively large values of n, such as $n = 1000$. Its optimal value is a lower bound on the optimal value of the two-way partitioning problem, and is always at least as good as the lower bound (5.8) based on $\lambda_{\min}(W)$.

5.2.3 Strong duality and Slater's constraint qualification

If the equality

$$d^\star = p^\star \tag{5.24}$$

holds, *i.e.*, the optimal duality gap is zero, then we say that *strong duality* holds. This means that the best bound that can be obtained from the Lagrange dual function is tight.

Strong duality does not, in general, hold. But if the primal problem (5.1) is convex, *i.e.*, of the form

$$\begin{array}{lll} \text{minimize} & f_0(x) & \\ \text{subject to} & f_i(x) \leq 0, & i = 1, \ldots, m, \\ & Ax = b, & \end{array} \tag{5.25}$$

with $f_0, \ldots, f_m$ convex, we usually (but not always) have strong duality. There are many results that establish conditions on the problem, beyond convexity, under which strong duality holds. These conditions are called *constraint qualifications*.

One simple constraint qualification is *Slater's condition*: There exists an $x \in \mathbf{relint}\,\mathcal{D}$ such that

$$f_i(x) < 0, \quad i = 1, \ldots, m, \qquad Ax = b. \tag{5.26}$$

Such a point is sometimes called *strictly feasible*, since the inequality constraints hold with strict inequalities. Slater's theorem states that strong duality holds, if Slater's condition holds (and the problem is convex).

Slater's condition can be refined when some of the inequality constraint functions f_i are affine. If the first k constraint functions $f_1, \ldots, f_k$ are affine, then strong duality holds provided the following weaker condition holds: There exists an $x \in \mathbf{relint}\,\mathcal{D}$ with

$$f_i(x) \leq 0, \quad i = 1, \ldots, k, \qquad f_i(x) < 0, \quad i = k+1, \ldots, m, \qquad Ax = b. \tag{5.27}$$

In other words, the affine inequalities do not need to hold with strict inequality. Note that the refined Slater condition (5.27) reduces to feasibility when the constraints are all linear equalities and inequalities, and $\mathbf{dom}\, f_0$ is open.

Slater's condition (and the refinement (5.27)) not only implies strong duality for convex problems. It also implies that the dual optimal value is attained when $d^\star > -\infty$, *i.e.*, there exists a dual feasible $(\lambda^\star, \nu^\star)$ with $g(\lambda^\star, \nu^\star) = d^\star = p^\star$. We will prove that strong duality obtains, when the primal problem is convex and Slater's condition holds, in §5.3.2.

5.2.4 Examples

Least-squares solution of linear equations

Recall the problem (5.5):

$$\begin{array}{ll} \text{minimize} & x^T x \\ \text{subject to} & Ax = b. \end{array}$$

The associated dual problem is

$$\text{maximize} \quad -(1/4)\nu^T A A^T \nu - b^T \nu,$$

which is an unconstrained concave quadratic maximization problem.

Slater's condition is simply that the primal problem is feasible, so $p^\star = d^\star$ provided $b \in \mathcal{R}(A)$, *i.e.*, $p^\star < \infty$. In fact for this problem we always have strong duality, even when $p^\star = \infty$. This is the case when $b \notin \mathcal{R}(A)$, so there is a z with $A^T z = 0$, $b^T z \neq 0$. It follows that the dual function is unbounded above along the line $\{tz \mid t \in \mathbf{R}\}$, so $d^\star = \infty$ as well.

Lagrange dual of LP

By the weaker form of Slater's condition, we find that strong duality holds for any LP (in standard or inequality form) provided the primal problem is feasible. Applying this result to the duals, we conclude that strong duality holds for LPs if the dual is feasible. This leaves only one possible situation in which strong duality for LPs can fail: both the primal and dual problems are infeasible. This pathological case can, in fact, occur; see exercise 5.23.

Lagrange dual of QCQP

We consider the QCQP

$$\begin{array}{ll} \text{minimize} & (1/2)x^T P_0 x + q_0^T x + r_0 \\ \text{subject to} & (1/2)x^T P_i x + q_i^T x + r_i \leq 0, \quad i = 1, \ldots, m, \end{array} \tag{5.28}$$

with $P_0 \in \mathbf{S}^n_{++}$, and $P_i \in \mathbf{S}^n_+$, $i = 1, \ldots, m$. The Lagrangian is

$$L(x, \lambda) = (1/2)x^T P(\lambda) x + q(\lambda)^T x + r(\lambda),$$

where

$$P(\lambda) = P_0 + \sum_{i=1}^m \lambda_i P_i, \qquad q(\lambda) = q_0 + \sum_{i=1}^m \lambda_i q_i, \qquad r(\lambda) = r_0 + \sum_{i=1}^m \lambda_i r_i.$$

It is possible to derive an expression for $g(\lambda)$ for general λ, but it is quite complicated. If $\lambda \succeq 0$, however, we have $P(\lambda) \succ 0$ and

$$g(\lambda) = \inf_x L(x, \lambda) = -(1/2)q(\lambda)^T P(\lambda)^{-1} q(\lambda) + r(\lambda).$$

We can therefore express the dual problem as

$$\begin{array}{ll} \text{maximize} & -(1/2)q(\lambda)^T P(\lambda)^{-1} q(\lambda) + r(\lambda) \\ \text{subject to} & \lambda \succeq 0. \end{array} \tag{5.29}$$

The Slater condition says that strong duality between (5.29) and (5.28) holds if the quadratic inequality constraints are strictly feasible, *i.e.*, there exists an x with

$$(1/2)x^T P_i x + q_i^T x + r_i < 0, \quad i = 1, \ldots, m.$$

Entropy maximization

Our next example is the entropy maximization problem (5.13):

$$\begin{array}{ll} \text{minimize} & \sum_{i=1}^n x_i \log x_i \\ \text{subject to} & Ax \preceq b \\ & \mathbf{1}^T x = 1, \end{array}$$

with domain $\mathcal{D} = \mathbf{R}^n_+$. The Lagrange dual function was derived on page 222; the dual problem is

$$\begin{array}{ll} \text{maximize} & -b^T\lambda - \nu - e^{-\nu-1} \sum_{i=1}^n e^{-a_i^T \lambda} \\ \text{subject to} & \lambda \succeq 0, \end{array} \tag{5.30}$$

with variables $\lambda \in \mathbf{R}^m$, $\nu \in \mathbf{R}$. The (weaker) Slater condition for (5.13) tells us that the optimal duality gap is zero if there exists an $x \succ 0$ with $Ax \preceq b$ and $\mathbf{1}^T x = 1$.

We can simplify the dual problem (5.30) by maximizing over the dual variable ν analytically. For fixed λ, the objective function is maximized when the derivative with respect to ν is zero, *i.e.*,

$$\nu = \log \sum_{i=1}^n e^{-a_i^T \lambda} - 1.$$

Substituting this optimal value of ν into the dual problem gives

$$\begin{array}{ll} \text{maximize} & -b^T\lambda - \log \left(\sum_{i=1}^n e^{-a_i^T \lambda} \right) \\ \text{subject to} & \lambda \succeq 0, \end{array}$$

which is a geometric program (in convex form) with nonnegativity constraints.

Minimum volume covering ellipsoid

We consider the problem (5.14):

$$\begin{array}{ll} \text{minimize} & \log \det X^{-1} \\ \text{subject to} & a_i^T X a_i \leq 1, \quad i = 1, \ldots, m, \end{array}$$

with domain $\mathcal{D} = \mathbf{S}^n_{++}$. The Lagrange dual function is given by (5.15), so the dual problem can be expressed as

$$\begin{array}{ll} \text{maximize} & \log\det\left(\sum_{i=1}^m \lambda_i a_i a_i^T\right) - \mathbf{1}^T\lambda + n \\ \text{subject to} & \lambda \succeq 0 \end{array} \tag{5.31}$$

where we take $\log\det X = -\infty$ if $X \not\succ 0$.

The (weaker) Slater condition for the problem (5.14) is that there exists an $X \in \mathbf{S}^n_{++}$ with $a_i^T X a_i \leq 1$, for $i = 1, \ldots, m$. This is always satisfied, so strong duality always obtains between (5.14) and the dual problem (5.31).

A nonconvex quadratic problem with strong duality

On rare occasions strong duality obtains for a *nonconvex* problem. As an important example, we consider the problem of minimizing a nonconvex quadratic function over the unit ball,

$$\begin{array}{ll} \text{minimize} & x^T A x + 2b^T x \\ \text{subject to} & x^T x \leq 1, \end{array} \tag{5.32}$$

where $A \in \mathbf{S}^n$, $A \not\succeq 0$, and $b \in \mathbf{R}^n$. Since $A \not\succeq 0$, this is not a convex problem. This problem is sometimes called the *trust region problem*, and arises in minimizing a second-order approximation of a function over the unit ball, which is the region in which the approximation is assumed to be approximately valid.

The Lagrangian is

$$L(x, \lambda) = x^T A x + 2b^T x + \lambda(x^T x - 1) = x^T(A + \lambda I)x + 2b^T x - \lambda,$$

so the dual function is given by

$$g(\lambda) = \begin{cases} -b^T(A + \lambda I)^\dagger b - \lambda & A + \lambda I \succeq 0, \quad b \in \mathcal{R}(A + \lambda I) \\ -\infty & \text{otherwise,} \end{cases}$$

where $(A + \lambda I)^\dagger$ is the pseudo-inverse of $A + \lambda I$. The Lagrange dual problem is thus

$$\begin{array}{ll} \text{maximize} & -b^T(A + \lambda I)^\dagger b - \lambda \\ \text{subject to} & A + \lambda I \succeq 0, \quad b \in \mathcal{R}(A + \lambda I), \end{array} \tag{5.33}$$

with variable $\lambda \in \mathbf{R}$. Although it is not obvious from this expression, this is a convex optimization problem. In fact, it is readily solved since it can be expressed as

$$\begin{array}{ll} \text{maximize} & -\sum_{i=1}^n (q_i^T b)^2/(\lambda_i + \lambda) - \lambda \\ \text{subject to} & \lambda \geq -\lambda_{\min}(A), \end{array}$$

where λ_i and q_i are the eigenvalues and corresponding (orthonormal) eigenvectors of A, and we interpret $(q_i^T b)^2/0$ as 0 if $q_i^T b = 0$ and as ∞ otherwise.

Despite the fact that the original problem (5.32) is not convex, we always have zero optimal duality gap for this problem: The optimal values of (5.32) and (5.33) are always the same. In fact, a more general result holds: strong duality holds for any optimization problem with quadratic objective and one quadratic inequality constraint, provided Slater's condition holds; see §B.1.

5.2.5 Mixed strategies for matrix games

In this section we use strong duality to derive a basic result for zero-sum matrix games. We consider a game with two players. Player 1 makes a choice (or *move*) $k \in \{1, \ldots, n\}$, and player 2 makes a choice $l \in \{1, \ldots, m\}$. Player 1 then makes a payment of P_{kl} to player 2, where $P \in \mathbf{R}^{n \times m}$ is the *payoff matrix* for the game. The goal of player 1 is to make the payment as small as possible, while the goal of player 2 is to maximize it.

The players use randomized or *mixed strategies*, which means that each player makes his or her choice randomly and independently of the other player's choice, according to a probability distribution:

$$\mathbf{prob}(k = i) = u_i, \quad i = 1, \ldots, n, \qquad \mathbf{prob}(l = i) = v_i, \quad i = 1, \ldots, m.$$

Here u and v give the probability distributions of the choices of the two players, *i.e.*, their associated strategies. The expected payoff from player 1 to player 2 is then

$$\sum_{k=1}^{n} \sum_{l=1}^{m} u_k v_l P_{kl} = u^T P v.$$

Player 1 wishes to choose u to minimize $u^T P v$, while player 2 wishes to choose v to maximize $u^T P v$.

Let us first analyze the game from the point of view of player 1, assuming her strategy u is known to player 2 (which clearly gives an advantage to player 2). Player 2 will choose v to maximize $u^T P v$, which results in the expected payoff

$$\sup\{u^T P v \mid v \succeq 0,\ \mathbf{1}^T v = 1\} = \max_{i=1,\ldots,m} (P^T u)_i.$$

The best thing player 1 can do is to choose u to minimize this worst-case payoff to player 2, *i.e.*, to choose a strategy u that solves the problem

$$\begin{array}{ll} \text{minimize} & \max_{i=1,\ldots,m} (P^T u)_i \\ \text{subject to} & u \succeq 0, \quad \mathbf{1}^T u = 1, \end{array} \tag{5.34}$$

which is a piecewise-linear convex optimization problem. We will denote the optimal value of this problem as $p_1^\star$. This is the smallest expected payoff player 1 can arrange to have, assuming that player 2 knows the strategy of player 1, and plays to his own maximum advantage.

In a similar way we can consider the situation in which v, the strategy of player 2, is known to player 1 (which gives an advantage to player 1). In this case player 1 chooses u to minimize $u^T P v$, which results in an expected payoff of

$$\inf\{u^T P v \mid u \succeq 0,\ \mathbf{1}^T u = 1\} = \min_{i=1,\ldots,n} (P v)_i.$$

Player 2 chooses v to maximize this, *i.e.*, chooses a strategy v that solves the problem

$$\begin{array}{ll} \text{maximize} & \min_{i=1,\ldots,n} (P v)_i \\ \text{subject to} & v \succeq 0, \quad \mathbf{1}^T v = 1, \end{array} \tag{5.35}$$

which is another convex optimization problem, with piecewise-linear (concave) objective. We will denote the optimal value of this problem as $p_2^\star$. This is the largest expected payoff player 2 can guarantee getting, assuming that player 1 knows the strategy of player 2.

It is intuitively obvious that knowing your opponent's strategy gives an advantage (or at least, cannot hurt), and indeed, it is easily shown that we always have $p_1^\star \geq p_2^\star$. We can interpret the difference, $p_1^\star - p_2^\star$, which is nonnegative, as the advantage conferred on a player by knowing the opponent's strategy.

Using duality, we can establish a result that is at first surprising: $p_1^\star = p_2^\star$. In other words, in a matrix game with mixed strategies, there is *no* advantage to knowing your opponent's strategy. We will establish this result by showing that the two problems (5.34) and (5.35) are Lagrange dual problems, for which strong duality obtains.

We start by formulating (5.34) as an LP,

$$\begin{array}{ll} \text{minimize} & t \\ \text{subject to} & u \succeq 0, \quad \mathbf{1}^T u = 1 \\ & P^T u \preceq t\mathbf{1}, \end{array}$$

with extra variable $t \in \mathbf{R}$. Introducing the multiplier λ for $P^T u \preceq t\mathbf{1}$, μ for $u \succeq 0$, and ν for $\mathbf{1}^T u = 1$, the Lagrangian is

$$t + \lambda^T(P^T u - t\mathbf{1}) - \mu^T u + \nu(1 - \mathbf{1}^T u) = \nu + (1 - \mathbf{1}^T\lambda)t + (P\lambda - \nu\mathbf{1} - \mu)^T u,$$

so the dual function is

$$g(\lambda, \mu, \nu) = \begin{cases} \nu & \mathbf{1}^T\lambda = 1, \quad P\lambda - \nu\mathbf{1} = \mu \\ -\infty & \text{otherwise.} \end{cases}$$

The dual problem is then

$$\begin{array}{ll} \text{maximize} & \nu \\ \text{subject to} & \lambda \succeq 0, \quad \mathbf{1}^T\lambda = 1, \quad \mu \succeq 0 \\ & P\lambda - \nu\mathbf{1} = \mu. \end{array}$$

Eliminating μ we obtain the following Lagrange dual of (5.34):

$$\begin{array}{ll} \text{maximize} & \nu \\ \text{subject to} & \lambda \succeq 0, \quad \mathbf{1}^T\lambda = 1 \\ & P\lambda \succeq \nu\mathbf{1}, \end{array}$$

with variables λ, ν. But this is clearly equivalent to (5.35). Since the LPs are feasible, we have strong duality; the optimal values of (5.34) and (5.35) are equal.